ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Electronic components and waveforms, motors and relay control circuits, hydraulic symbols and circuits, electro-hydraulic control, and fault-finding and maintenance.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Use only a black or blue pen.
6. Write neatly and legibly.

QUESTION 1

1.1 Define the term *diode*. (2)

1.2 List TWO applications of a diode. (2)

1.3 Differentiate between an *analogue signal* and a *digital signal*. (2 × 2) (4)

1.4 FIGURE 1 below shows the waveform of an alternating voltage.

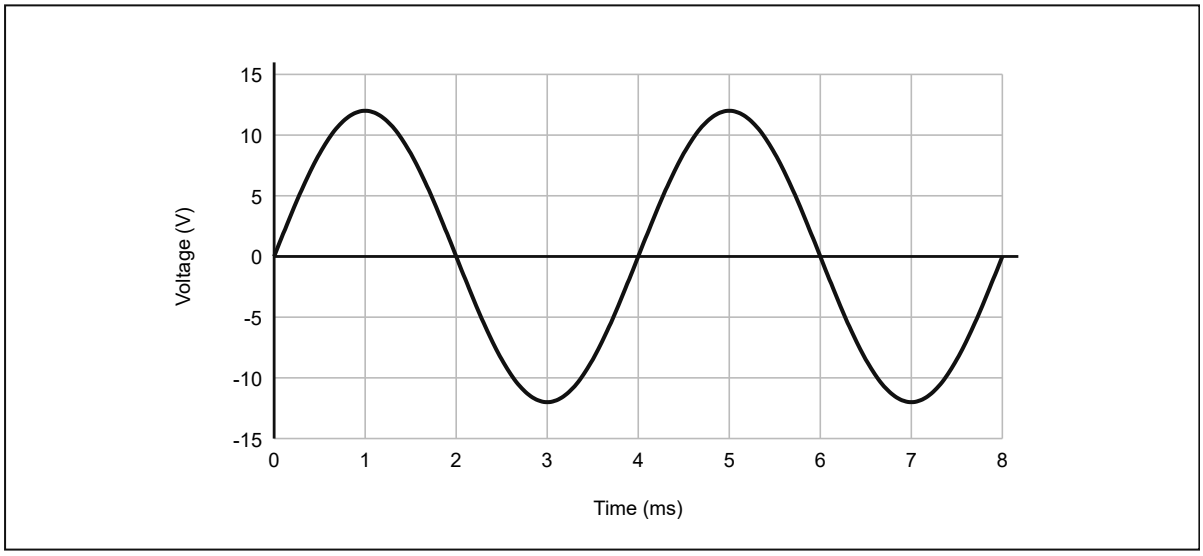


FIGURE 1

Use FIGURE 1 to determine the following:

1.4.1 The peak value of the voltage (1)

1.4.2 The peak-to-peak value of the voltage (1)

1.4.3 The period of the waveform (2)

1.4.4 The frequency of the waveform (show ALL calculations) (3)

1.5 Draw a fully labelled symbol of a light-emitting diode (LED). Show the direction in which current flows through it. (5)

[20]

QUESTION 2

2.1 Define the term *stepper motor*. (2)

2.2 Name THREE methods of controlling the speed of a DC motor. (3)

2.3 Explain the function of a limit switch in a control circuit. (2)

2.4 State THREE advantages of using a relay to switch a motor. (3)

2.5 FIGURE 2 below shows the start/stop control circuit of a DC motor.

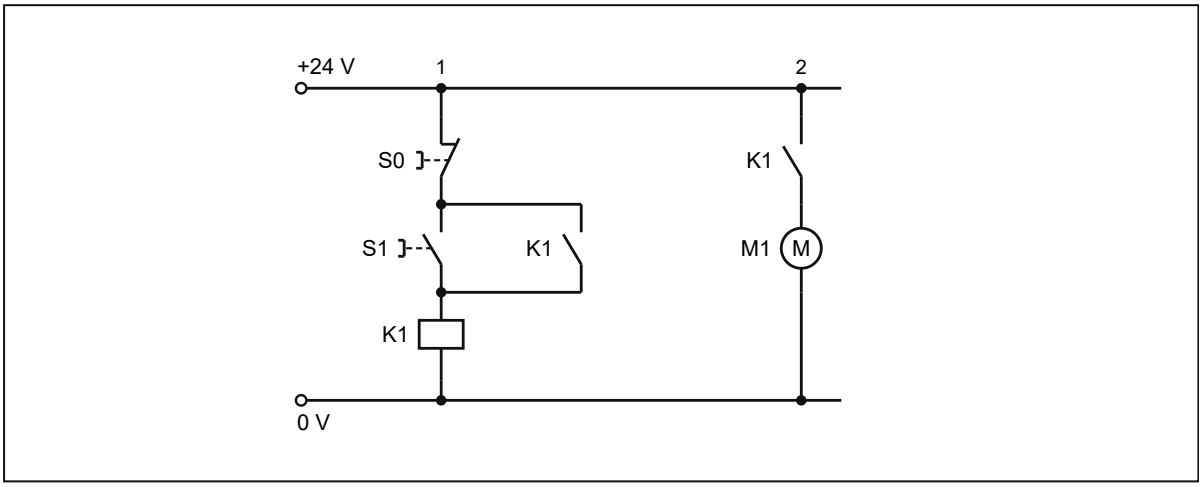


FIGURE 2

Redraw the circuit and add the following:

• An emergency stop push button (S2) that stops the motor
• A second STOP push button (S3) so that the motor can also be stopped from another place
• A pilot lamp (H1) that is on while the motor runs

(10)

[20]

QUESTION 3

3.1 Draw the symbol of EACH of the following:

3.1.1 Fixed-displacement hydraulic pump with one direction of flow (2)

3.1.2 Non-return (check) valve (2)

3.1.3 Pressure gauge (2)

3.2 Label the parts of the hydraulic circuit shown in FIGURE 3 below. Write only the answer next to the question number (3.2.1–3.2.6) in the ANSWER BOOK.

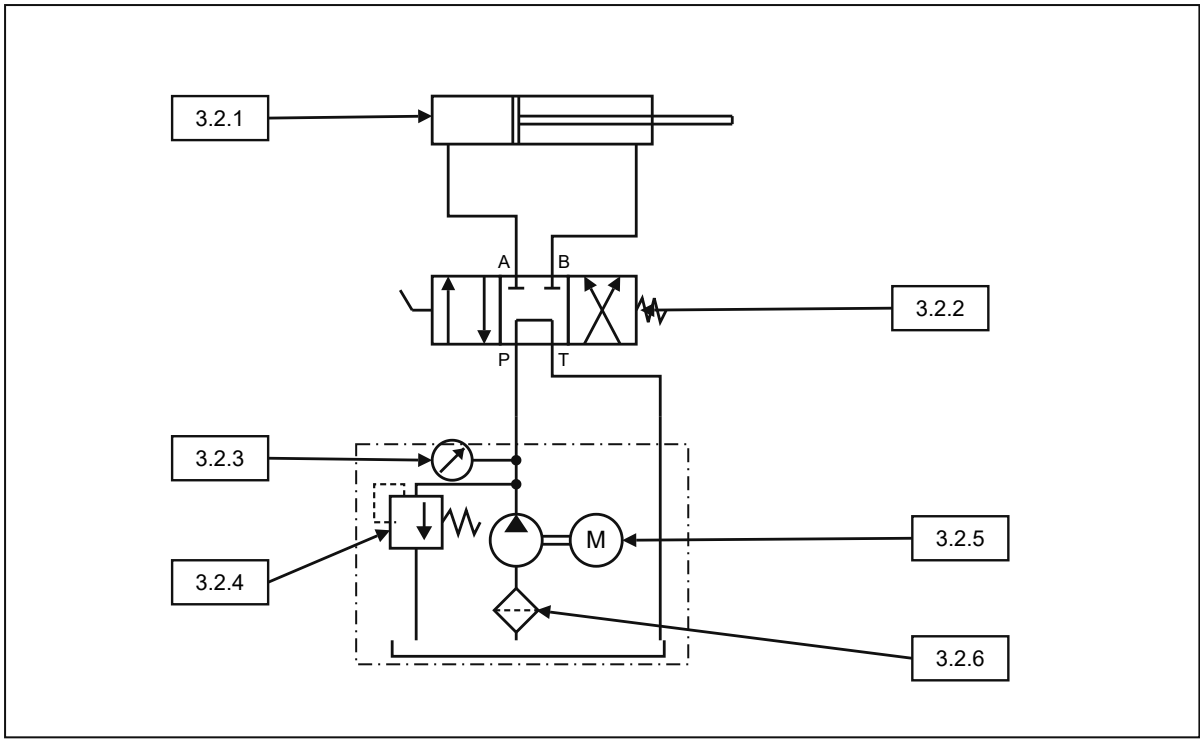


FIGURE 3

(6 × 1) (6)

3.3 Design a hydraulic circuit to control the speed of a double-acting cylinder while it extends (meter-out control). Use a manually operated 4/3-way valve with a tandem centre, a one-way flow control valve and a hydraulic power pack. (8)

[20]

QUESTION 4

4.1 Explain the function of the solenoid on an electro-hydraulic directional control valve. (2)

4.2 Differentiate between a *normally open (NO)* and a *normally closed (NC)* push button. (4)

(2 × 2)

4.3 Design the electro-hydraulic circuit (hydraulic working circuit and electrical control circuit) for the following: A double-acting cylinder is controlled by a 4/2-way valve with a single solenoid (1M1) and a spring return. When the START push button (S1) is pressed, the cylinder extends and relay K1 holds itself in. When the cylinder reaches the end of its stroke, limit switch S2 releases K1 and the cylinder retracts. (12)

4.4 Explain what is meant by the term *interlocking* in a control circuit. (2)

[20]

QUESTION 5

5.1 Explain the purpose of preventive maintenance. (2)

5.2 Explain the procedure to follow when testing a relay coil for continuity with a multimeter. (5)

5.3 A hydraulic cylinder slowly moves (creeps) when the directional control valve is in the centre position. Briefly explain the action you would take for EACH of the following possible causes:

5.3.1 Worn piston seals in the cylinder (2)

5.3.2 A leaking directional control valve spool (2)

5.3.3 Air in the hydraulic system (2)

5.4 Explain the procedure to follow when changing the oil of a hydraulic system. (7)

[20]

TOTAL: 100

END OF QUESTION PAPER